$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & -\frac{Y_1 k^2}{2} & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{1}{2}(Y_2 k^2 + \overset{(2)}{\kappa}_1) & \frac{-Y_3 k^2 - 2\overset{(2)}{\kappa}_1}{2\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{-Y_3 k^2 - 2\overset{(2)}{\kappa}_1}{2\sqrt{2}} & \frac{1}{2}(Y_4 k^2 + 2\overset{(2)}{\kappa}_1) & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{1}{2}(Y_5 k^2 + \overset{(2)}{\kappa}_1) & \frac{Y_5 k^2 + \overset{(2)}{\kappa}_1}{\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{Y_5 k^2 + \overset{(2)}{\kappa}_1}{\sqrt{2}} & Y_5 k^2 + \overset{(2)}{\kappa}_1
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{1}{2}(Y_6 k^2 + 3\overset{(2)}{\kappa}_1) & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{2}(3 k^2 \overset{(4)}{\kappa}_{15} + 3\overset{(2)}{\kappa}_1)
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{\mathrm{Det}(0^+)} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{Y_4 k^2 + 2\overset{(2)}{\kappa}_1}{2\,\mathrm{Det}(1^+)} & \frac{Y_3 k^2 + 2\overset{(2)}{\kappa}_1}{2\sqrt{2}\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{Y_3 k^2 + 2\overset{(2)}{\kappa}_1}{2\sqrt{2}\,\mathrm{Det}(1^+)} & \frac{Y_2 k^2 + \overset{(2)}{\kappa}_1}{2\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{1}{3\,\mathrm{Det}(1^-)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{2}{3\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{\mathrm{Det}(2^-)}
\end{array}
$$

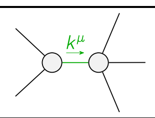
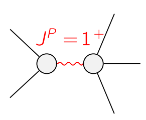
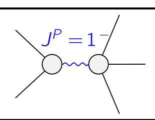
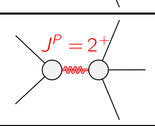
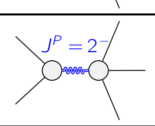
Abbreviations used in matrices

$$
\begin{aligned}
&Y_1 == 3\overset{(4)}{\kappa}_1 - 3\overset{(4)}{\kappa}_{15} + 3\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \;\&\&\; Y_2 == \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \;\&\& \\
&Y_3 == 2\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_4 \;\&\&\; Y_4 == 11\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \;\&\&\; Y_5 == 3\overset{(4)}{\kappa}_{15} - 2\overset{(4)}{\kappa}_2 \;\&\& \\
&Y_6 == 3\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 \;\&\&\; \mathrm{Det}(0^+) == -\frac{1}{2} k^2 (3\overset{(4)}{\kappa}_1 - 3\overset{(4)}{\kappa}_{15} + 3\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) \;\&\& \\
&\mathrm{Det}(1^+) == \frac{1}{8} (k^4 (18\overset{(4)}{\kappa}_{15}{}^2 - 18\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_2 - 20\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_3 + 18\overset{(4)}{\kappa}_2\overset{(4)}{\kappa}_3 - \\
&\quad 2\overset{(4)}{\kappa}_3{}^2 + 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_4 - 2\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2 + 2(11\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)\overset{(4)}{\kappa}_6) + \\
&\quad 2k^2 (9\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 + 3\overset{(4)}{\kappa}_4 + 2\overset{(4)}{\kappa}_6)\overset{(2)}{\kappa}_1) \;\&\&\; \mathrm{Det}(2^+) == \frac{1}{2} (k^2 (3\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6) + 3\overset{(2)}{\kappa}_1)
\end{aligned}
$$

Lagrangian

$$
\begin{aligned}
&\overset{(2)}{\kappa}_1 \mathcal{K}_{\alpha\beta\chi}\mathcal{K}^{\alpha\beta\chi} + \overset{(2)}{\kappa}_1 \mathcal{K}^{\alpha\beta\chi}\mathcal{K}_{\beta\alpha\chi} + \overset{(2)}{\kappa}_1 \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\mathcal{K}^{\chi}{}_{\beta\chi} + \overset{(4)}{\kappa}_1 \partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \\
&\overset{(4)}{\kappa}_2 \partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3 \partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_4 \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \overset{(4)}{\kappa}_6 \partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \\
&6\overset{(4)}{\kappa}_{15} \partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - 6\overset{(4)}{\kappa}_2 \partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} + \frac{9}{2}\overset{(4)}{\kappa}_{15} \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \frac{9}{2}\overset{(4)}{\kappa}_2 \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \\
&\frac{1}{2}\overset{(4)}{\kappa}_3 \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\overset{(4)}{\kappa}_4 \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta\mathcal{K}_{\alpha\beta\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta\mathcal{K}_{\beta\alpha\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi}
\end{aligned}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^-}{}^{\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1}_{1^-}{}^{\alpha} == \mathcal{J}^{\#2}_{1^-}{}^{\alpha}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
Total # constraint(s):	4	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{(9\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 + 3\overset{(4)}{\kappa}_4 + 2\overset{(4)}{\kappa}_6)(108\overset{(2)}{\kappa}_1{}^2 + 7(18\overset{(4)}{\kappa}_{15} - 18\overset{(4)}{\kappa}_2 + 4\overset{(4)}{\kappa}_3 + 3\overset{(4)}{\kappa}_4 - 2\overset{(4)}{\kappa}_6)^2 p^4)}{(-9\overset{(4)}{\kappa}_{15} + 9\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 - 3\overset{(4)}{\kappa}_4 - 2\overset{(4)}{\kappa}_6)^2\overset{(2)}{\kappa}_1{}^2}$
	3	$\frac{2(-9\overset{(4)}{\kappa}_{15} + 9\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 - 3\overset{(4)}{\kappa}_4 - 2\overset{(4)}{\kappa}_6)\overset{(2)}{\kappa}_1}{18\overset{(4)}{\kappa}_{15}{}^2 + 18\overset{(4)}{\kappa}_2\overset{(4)}{\kappa}_3 - 2\overset{(4)}{\kappa}_3{}^2 - 2\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4{}^2 + 2(-9\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)\overset{(4)}{\kappa}_6 + \overset{(4)}{\kappa}_{15}(-18\overset{(4)}{\kappa}_2 - 20\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4 + 22\overset{(4)}{\kappa}_6)}$	$(2(162\overset{(4)}{\kappa}_{15}{}^2 + 162\overset{(4)}{\kappa}_2{}^2 + 6\overset{(4)}{\kappa}_3{}^2 - 36\overset{(4)}{\kappa}_{15}(9\overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 - 2\overset{(4)}{\kappa}_4) +$ $4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 + 9\overset{(4)}{\kappa}_4{}^2 - 36\overset{(4)}{\kappa}_2(\overset{(4)}{\kappa}_3 + 2\overset{(4)}{\kappa}_4) - 8\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_6 + 4\overset{(4)}{\kappa}_4\overset{(4)}{\kappa}_6 + 4\overset{(4)}{\kappa}_6{}^2))/$ $(162\overset{(4)}{\kappa}_{15}{}^3 + 2\overset{(4)}{\kappa}_3{}^3 - 4\overset{(4)}{\kappa}_3{}^2\overset{(4)}{\kappa}_4 - 5\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4{}^2 - 3\overset{(4)}{\kappa}_4{}^3 - 18\overset{(4)}{\kappa}_{15}{}^2(18\overset{(4)}{\kappa}_2 + 11\overset{(4)}{\kappa}_3 - 6\overset{(4)}{\kappa}_4 - 13\overset{(4)}{\kappa}_6) -$ $162\overset{(4)}{\kappa}_2{}^2(\overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6) - 6\overset{(4)}{\kappa}_3{}^2\overset{(4)}{\kappa}_6 + 4\overset{(4)}{\kappa}_4{}^2\overset{(4)}{\kappa}_6 + 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_6{}^2 + 4\overset{(4)}{\kappa}_4\overset{(4)}{\kappa}_6{}^2 +$ $9\overset{(4)}{\kappa}_2(8\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2 + 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_6 - 8\overset{(4)}{\kappa}_4\overset{(4)}{\kappa}_6 - 4\overset{(4)}{\kappa}_6{}^2) + \overset{(4)}{\kappa}_{15}$ $(162\overset{(4)}{\kappa}_2{}^2 + 2\overset{(4)}{\kappa}_3{}^2 - 84\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 + 9\overset{(4)}{\kappa}_4{}^2 + 36\overset{(4)}{\kappa}_2(10\overset{(4)}{\kappa}_3 - 3\overset{(4)}{\kappa}_4 - 11\overset{(4)}{\kappa}_6) - 44\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_6 + 96\overset{(4)}{\kappa}_4\overset{(4)}{\kappa}_6 + 44\overset{(4)}{\kappa}_6{}^2))$
	3	$\frac{\overset{(2)}{\kappa}_1}{-3\overset{(4)}{\kappa}_{15} + 2\overset{(4)}{\kappa}_2}$	$-\frac{2}{9\overset{(4)}{\kappa}_{15} - 6\overset{(4)}{\kappa}_2}$
	5	$\frac{3\overset{(2)}{\kappa}_1}{-3\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6}$	$-\frac{2}{-3\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6}$
	5	$-\frac{\overset{(2)}{\kappa}_1}{\overset{(4)}{\kappa}_{15}}$	$-\frac{2}{3\overset{(4)}{\kappa}_{15}}$
Resolved unitarity condition(s):		(Demonstrably impossible)	